

TAFSIR CHOWDHURY

Full Stack AI Engineer

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SUMMARY

Backend-first Full Stack AI Engineer with system design depth, AI pipeline ownership, and DevOps working knowledge. 3+ years building production-grade web applications with TypeScript, Node.js, Next.js, and cloud infrastructure (AWS EC2/ECS, Railway). Proven track record architecting multi-LLM pipelines (OpenAI, Claude, Gemini), distributed worker systems, and scalable REST APIs for global clients. BSc in Computer Science with a 3.67 GPA and multiple IBM and Meta certifications.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Tanstack Query, Zustand, Redux, Tailwind CSS, ShadCN UI, Ant Design

Backend: Node.js, Express.js, Drizzle, Prisma ORM, Mongoose, Redis, BullMQ, Better Auth, Auth.js, Clerk

AI & ML: LangChain, LangGraph, Vercel AI SDK, OpenAI API, Claude API, Gemini API, LangSmith

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Neon.tech

DevOps & Cloud: AWS (EC2, ECS), Docker, GitHub Actions, DigitalOcean, Vercel, CI/CD Pipelines

Tools: Git, GitHub, Postman, Linux

PROFESSIONAL EXPERIENCE

Standard Insights

Full Stack AI Engineer

Oct '24 – Present

Remote | New York, USA

- Architected and shipped a **multi-phase, multi-LLM AI report generation pipeline** powering market trend analysis, consumer persona synthesis, and strategic insights – built as **32 durable Inngest functions** with automatic retries, concurrency control, and failure callbacks, replacing a brittle single-request model and eliminating job loss on server restarts.
- Designed and owned a **distributed background worker microservice** processing AI workloads across **10+ concurrent queue types** (quantitative data analysis, interview translation, persona generation, B2B scraping, Google Sheets sync); implemented batching and per-survey concurrency locks to prevent race conditions at scale.
- Built an **AI-powered Survey Builder Engine** with 7-intent discriminated-union Zod schema validation against Gemini structured output, enabling natural-language survey authoring and real-time LLM-to-state mutation with zero schema-drift between the API and database layer.
- Engineered a **6-step autonomous AI agent pipeline** that ingests user-uploaded Excel/CSV files from AWS S3, validates format, parses multi-sheet structures, runs LLM-based schema analysis via Gemini, extracts survey questions and respondents, and re-uploads a normalized output file - all as idempotent, memoized steps with 3-retry fault tolerance.
- Built and optimized **RESTful APIs across 50+ Express modules** using MongoDB (50+ Mongoose models) and Redis caching, improving session state management and data-layer response consistency; implemented JWT-based auth, role-based access control, and Google OAuth2 token refresh lifecycle for third-party integrations (Google Sheets, Slack).
- Containerized **3 production services** (client, server, admin) with Docker and managed deployments on **AWS ECS, Railway**, integrating CI/CD pipelines via GitHub Actions; deployed the worker service as an **AWS Lambda** function for cost-optimized serverless execution at scale.
- Delivered consumer intelligence dashboards and survey creation UIs using **Next.js 15 App Router, TypeScript, TanStack Query, Zustand, and ShadCN/Radix UI**, contributing to a **Turborepo monorepo** spanning **4 apps and 19 shared packages**, reducing cross-app code duplication by an estimated **~40%**.

HJ Animation Shop LTD

Full-Stack Developer

Jul '23 – Sep '24

Remote | Loughton, United Kingdom

- Built **autonomous AI customer support platform (Aximus)** using **Next.js and Vercel AI SDK** – engineered a **ToolLoopAgent** with dynamic tool-calling and a **keyword-scoring RAG engine** (no embedding API; **98% accuracy**), with **HMAC-SHA256 Meta webhook ingestion** and atomic DB idempotency locks.
- Architected **multi-tenant SaaS platforms** in **Next.js 16 and Drizzle ORM**, including a **floating-point-free commission math engine**, custom binary **JPEG/PNG metadata parser** for AI portrait editing, and randomized exam engine with practical video submission grading.
- Integrated **Better Auth** with **RBAC, 6-digit OTP 2FA**, and session revocation across products; wired **SumUp and UddoktaPay v2** payment flows, and serverless **jsPDF** routes with font-scaling for dynamic certificate generation.
- Delivered **headless CMS (Parocia)** with **React Server Components** (zero client JS on public pages), dynamic sitemap/robots.txt/JSON-LD schema compilation, **UUID-hashed file storage** with path-traversal protection.

PROJECTS

Visual Source | *Real-time soccer match operations and social graphic automation*

Live | 2026

- Architected a multi-tenant soccer club dashboard using **Next.js 16**, **Turborepo**, and **Better-Auth**, securing isolated workspaces for multiple sports organizations.
- Engineered a headless canvas rendering pipeline with **Fabric.js**, **node-canvas**, and **Sharp** to compile dynamic matchday graphics stored in **Cloudflare R2**.
- Integrated **Facebook**, **Instagram**, and **Twitter Graph APIs**, executing a polling and media chunking pipeline for real-time automated matchday posting.
- Designed a resilient **Sportmonks API** data synchronizer using **PostgreSQL** advisory locks to prevent database write collisions during concurrent syncing.
- Developed a live event logging UI with game-state tracking, automatic score override mechanisms, and a cron-scheduled graphic publishing queue.
- Deployed a containerized production stack using **Docker Compose**, orchestrating PostgreSQL, Next.js, and Caddy to automate SSL/HTTPS certificates and reverse proxying.

Everything Green | *Digital Sustainability & SEO Platform*

Live | 2025

- Designed a **Retrieval-Augmented Generation (RAG)** chatbot using **MongoDB** and **OpenAI** embeddings, featuring dynamic HTML scraping, semantic chunking, and context-aware query optimization to deliver digital sustainability advice.
- Developed an **AI SEO recommendation engine** that matches **Google Ads** suggestions to scraped website pages via cosine similarity, classifying conversion performance using a custom-trained **CatBoost** model.
- Implemented a secure subscription billing system using the **Factory Pattern** to integrate **Stripe** and **PayPal**, supporting customer portals, dynamic coupons, and webhook validation.
- Implemented a dynamic website badge generation system using **Puppeteer**-based headless rendering, dynamically generating customized carbon certification badges from HTML templates and real-time site scores.
- Built a responsive **Next.js 15** dashboard with interactive **D3/Sankey** data visualizations and developed a **Manifest V3 Chrome Extension** to inject performance auditing metrics directly into Google SERPs.

Career Dock | *A Modern Job tracking application*

Live | 2024

- Built an end-to-end **Next.js** and **Express.js** web application utilizing **Redux Toolkit (RTK Query)** for cached state management, reducing client-side API latency by **30%**.
- Integrated the **Gemini API** to construct prompt-chaining workflows that generate tailored resumes, LinkedIn profiles, and cover letters based on target job descriptions.
- Architected secure authentication and subscription monetization pipelines using **Clerk** for multi-tenant identity verification and **Lemon Squeezy** for premium plan billing.
- Designed interactive analytical dashboards using real-time charting components powered by dynamic **Mongoose** aggregations to track job application success rates.
- Deployed the distributed architecture to **Vercel** and cloud servers with a fully automated **CI/CD pipeline**, ensuring zero-downtime releases and instant preview deployments.

EDUCATION

University of the People

2021 – 2024

Bachelor of Science in Computer Science

DEAC | California, USA

- Graduated with CGPA 3.67
- Enlisted for Dean's Award List

CERTIFICATIONS

IBM: Full Stack Software Developer Specialization

Meta: Front-End Developer Specialization | Back-End Developer Specialization

Google: IT Support Specialization

Amazon Web Services: AWS Fundamentals Specialization

LANGUAGES

English: (Comfortable - professional and technical communication)

Bengali: (Native)

Hindi: (Communicative)